



spring boot

Logging

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Logging is the process of *capturing events* and *messages* that occur during the execution of a program, and storing them for later analysis. It provides a *means of debugging, troubleshooting, and understanding the behavior of an application*.

Logging It can help *identify errors, performance issues, security breaches, and other problems that may occur during runtime*.

Logging Libraries and Frameworks for Java

Log4j: It is a powerful and flexible logging framework that supports various logging levels and appenders.

Logback: It is the successor of **Log4j** and is designed to be faster, more reliable, and more flexible than its predecessor.

SLF4J (Simple Logging Facade for Java): It is a logging facade or abstraction that provides a common API for various logging frameworks such as **Log4j2**, **Logback**

Apache Commons Logging: It is a thin layer on top of other logging frameworks that provides a common API for logging.

Log4j2: It is the successor of Log4j and provides better performance, scalability, and reliability than its predecessor.

Logging in Spring boot

Spring Boot provides a powerful and flexible logging framework based on the popular **SLF4J (Simple Logging Facade for Java) API**. This logging framework is highly customizable and can be configured to use a variety of logging implementations such as **Logback**, **Log4j2**, and **Java Util Logging (JUL)**.

- **ERROR**
- **WARN**
- **INFO**
- **DEBUG**
- **TRACE**

Request and Response Logging

Logbook is an extensible Java library to enable complete request and response logging for different **client-side** and **server-side** technologies. It satisfies a special need by a) allowing web application developers to log any HTTP traffic that an application receives or sends b) in a way that makes it easy to persist and analyze it later. This can be useful for traditional log analysis, meeting audit requirements or investigating individual historic traffic issues.

Logbook is ready to use out of the box for most common setups.

```
<dependency>  
  <groupId>org.zalando</groupId>  
  <artifactId>logbook-spring-boot-starter</artifactId>  
  <version>3.0.0-RC.0</version>  
</dependency>
```

Interview Questions